

Signature of the Invigilator



**NOTATIONS USED**

**R** : The set of all real numbers

**Z** : The set of all integers

**N** : The set of all natural numbers 1, 2, 3, ...

$$i = \sqrt{-1}$$

DO NOT WRITE ON THIS PAGE



**IMPORTANT NOTE FOR CANDIDATES**

- Attempt ALL the 29 questions.
- Questions 1-15 (objective questions) carry six marks each and questions 16-29 (subjective questions) carry fifteen marks each.
- Write the answers to the objective questions in the Answer Table for Objective Questions provided on page 7 only.

1. Let  $A(t)$  denote the area bounded by the curve  $y=e^{-|x|}$ , the  $x$ -axis and the straight lines  $x=-t$  and  $x=t$ . Then  $\lim_{t \rightarrow \infty} A(t)$  is equal to
  - (A) 2
  - (B) 1
  - (C)  $1/2$
  - (D) 0
  
2. If  $k$  is a constant such that  $xy+k=e^{(x-1)^2/2}$  satisfies the differential equation  $x \frac{dy}{dx} = (x^2 - x - 1)y + (x - 1)$ , then  $k$  is equal to
  - (A) 1
  - (B) 0
  - (C) -1
  - (D) -2
  
3. Which of the following functions is uniformly continuous on the domain as stated?
  - (A)  $f(x) = x^2, \quad x \in \mathbf{R}$
  - (B)  $f(x) = \frac{1}{x}, \quad x \in [1, \infty)$
  - (C)  $f(x) = \tan x, \quad x \in (-\pi/2, \pi/2)$
  - (D)  $f(x) = [x], \quad x \in [0, 1]$   
 ( $[x]$  is the greatest integer less than or equal to  $x$ )



4. Let  $R$  be the ring of polynomials over  $\mathbf{Z}_2$  and let  $I$  be the ideal of  $R$  generated by the polynomial  $x^3 + x + 1$ . Then the number of elements in the quotient ring  $R/I$  is

(A) 2  
(B) 4  
(C) 8  
(D) 16

5. Which of the following sets is a basis for the subspace

$$W = \left\{ \begin{bmatrix} x & y \\ 0 & t \end{bmatrix} : x + 2y + t = 0, y + t = 0 \right\}$$

of the vector space of all real  $2 \times 2$  matrices?

(A)  $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

(B)  $\left\{ \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \right\}$

(C)  $\left\{ \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix} \right\}$

(D)  $\left\{ \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \right\}$

6. Let  $G$  be an Abelian group of order 10. Let  $S = \{g \in G : g^{-1} = g\}$ . Then the number of non-identity elements in  $S$  is

(A) 5  
(B) 2  
(C) 1  
(D) 0

---

Space for rough work



7. Let  $(a_n)$  be an increasing sequence of positive real numbers such that the series  $\sum_{k=1}^{\infty} a_k$  is divergent. Let  $s_n = \sum_{k=1}^n a_k$  for  $n = 1, 2, \dots$  and  $t_n = \sum_{k=2}^n \frac{a_k}{s_{k-1}s_k}$  for  $n = 2, 3, \dots$ . Then  $\lim_{n \rightarrow \infty} t_n$  is equal to

- (A)  $1/a_1$
- (B)  $0$
- (C)  $1/(a_1 + a_2)$
- (D)  $a_1 + a_2$

8. For every function  $f : [0, 1] \rightarrow \mathbf{R}$  which is twice differentiable and satisfies  $f'(x) \geq 1$  for all  $x \in [0, 1]$ , we must have

- (A)  $f''(x) \geq 0$  for all  $x \in [0, 1]$
- (B)  $f(x) \geq x$  for all  $x \in [0, 1]$
- (C)  $f(x_2) - x_2 \leq f(x_1) - x_1$  for all  $x_1, x_2 \in [0, 1]$  with  $x_2 \geq x_1$
- (D)  $f(x_2) - x_2 \geq f(x_1) - x_1$  for all  $x_1, x_2 \in [0, 1]$  with  $x_2 \geq x_1$

9. Let  $f : \mathbf{R}^2 \rightarrow \mathbf{R}$  be defined by

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Which of the following statements holds regarding the continuity and the existence of partial derivatives of  $f$  at  $(0, 0)$ ?

- (A) Both partial derivatives of  $f$  exist at  $(0, 0)$  and  $f$  is continuous at  $(0, 0)$
- (B) Both partial derivatives of  $f$  exist at  $(0, 0)$  and  $f$  is NOT continuous at  $(0, 0)$
- (C) One partial derivative of  $f$  does NOT exist at  $(0, 0)$  and  $f$  is continuous at  $(0, 0)$
- (D) One partial derivative of  $f$  does NOT exist at  $(0, 0)$  and  $f$  is NOT continuous at  $(0, 0)$

Space for rough work

10. Suppose  $(c_n)$  is a sequence of real numbers such that  $\lim_{n \rightarrow \infty} |c_n|^{1/n}$  exists and is non-zero.

If the radius of convergence of the power series  $\sum_{n=0}^{\infty} c_n x^n$  is equal to  $r$ , then the radius of

convergence of the power series  $\sum_{n=1}^{\infty} n^2 c_n x^n$  is

- (A) less than  $r$
- (B) greater than  $r$
- (C) equal to  $r$
- (D) equal to 0

11. The rank of the matrix  $\begin{bmatrix} 1 & 4 & 8 \\ 2 & 10 & 22 \\ 0 & 4 & 12 \end{bmatrix}$  is

- (A) 3
- (B) 2
- (C) 1
- (D) 0

12. Let  $f: \mathbf{R} \rightarrow \mathbf{R}$  be a continuous function. If  $\int_0^x f(2t) dt = \frac{x}{\pi} \sin(\pi x)$  for all  $x \in \mathbf{R}$ , then  $f(2)$  is equal to

- (A) -1
- (B) 0
- (C) 1
- (D) 2

---

Space for rough work



13. Let  $\vec{u} = (ae^x \sin y - 4x)\hat{i} + (2y + e^x \cos y)\hat{j} + az\hat{k}$ , where  $a$  is a constant. If the line integral  $\oint_C \vec{u} \cdot d\vec{r}$  over every closed curve  $C$  is zero, then  $a$  is equal to
- (A)  $-2$   
(B)  $-1$   
(C)  $0$   
(D)  $1$
14. One of the integrating factors of the differential equation  $(y^2 - 3xy)dx + (x^2 - xy)dy = 0$  is
- (A)  $1/(x^2y^2)$   
(B)  $1/(x^2y)$   
(C)  $1/(xy^2)$   
(D)  $1/(xy)$
15. Let  $C$  denote the boundary of the semi-circular disk  $D = \{(x, y) \in \mathbf{R}^2 : x^2 + y^2 \leq 1, y \geq 0\}$  and let  $\phi(x, y) = x^2 + y$  for  $(x, y) \in D$ . If  $\hat{n}$  is the outward unit normal to  $C$ , then the integral  $\oint_C (\vec{\nabla} \phi) \cdot \hat{n} ds$ , evaluated counter-clockwise over  $C$ , is equal to
- (A)  $0$   
(B)  $\pi - 2$   
(C)  $\pi$   
(D)  $\pi + 2$

---

Space for rough work



### *Answer Table for Objective Questions*

Write the Code of your chosen answer only in the 'Answer' column against each Question No. Do not write anything else on this page:

| Question No. | Answer | Do not write in this column |
|--------------|--------|-----------------------------|
| 01           |        |                             |
| 02           |        |                             |
| 03           |        |                             |
| 04           |        |                             |
| 05           |        |                             |
| 06           |        |                             |
| 07           |        |                             |
| 08           |        |                             |
| 09           |        |                             |
| 10           |        |                             |
| 11           |        |                             |
| 12           |        |                             |
| 13           |        |                             |
| 14           |        |                             |
| 15           |        |                             |

#### FOR EVALUATION ONLY

|                                   |  |       |       |
|-----------------------------------|--|-------|-------|
| No. of Correct Answers            |  | Marks | ( + ) |
| No. of Incorrect Answers          |  | Marks | ( - ) |
| Total Marks in Question Nos. 1-15 |  |       | ( )   |



|          |
|----------|
| <b>B</b> |
|----------|

16. (a) Let  $M = \begin{bmatrix} 1+i & 2i & i+3 \\ 0 & 1-i & 3i \\ 0 & 0 & i \end{bmatrix}$ .

Determine the eigenvalues of the matrix

$$B = M^2 - 2M + I. \quad (9)$$

- (b) Let  $N$  be a square matrix of order 2. If the determinant of  $N$  is equal to 9 and the sum of the diagonal entries of  $N$  is equal to 10, then determine the eigenvalues of  $N$ . (6)



17. (a) Using the method of variation of parameters, solve the differential equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x^2,$$

given that  $x$  and  $\frac{1}{x}$  are two solutions of the corresponding homogeneous equation.

(9)

- (b) Find the real number  $\alpha$  such that the differential equation

$$\frac{d^2 y}{dx^2} + 2(\alpha - 1)(\alpha - 3) \frac{dy}{dx} + (\alpha - 2)y = 0$$

has a solution  $y(x) = a \cos(\beta x) + b \sin(\beta x)$  for some non-zero real numbers  $a, b, \beta$ .

(6)



18. (a) Let  $a, b, c$  be non-zero real numbers such that  $(a-b)^2 = 4ac$ . Solve the differential

$$\text{equation } a(x + \sqrt{2})^2 \frac{d^2 y}{dx^2} + b(x + \sqrt{2}) \frac{dy}{dx} + cy = 0. \quad (9)$$

(b) Solve the differential equation

$$dx + (e^{y \sin y} - x)(y \cos y + \sin y) dy = 0. \quad (6)$$



19. Let  $f(x, y) = x(x - 2y^2)$  for  $(x, y) \in \mathbf{R}^2$ . Show that  $f$  has a local minimum at  $(0, 0)$  on every straight line through  $(0, 0)$ . Is  $(0, 0)$  a critical point of  $f$ ? Find the discriminant of  $f$  at  $(0, 0)$ . Does  $f$  have a local minimum at  $(0, 0)$ ? Justify your answers. (15)



20. (a) Find the finite volume enclosed by the paraboloids  $z = 2 - x^2 - y^2$  and  $z = x^2 + y^2$ . (9)

(b) Let  $f : [0, 3] \rightarrow \mathbf{R}$  be a continuous function with  $\int_0^3 f(x) dx = 3$ . Evaluate

$$\int_0^3 [xf(x) + \int_0^x f(t) dt] dx. \quad (6)$$

21. (a) Let  $S$  be the surface  $\{(x, y, z) \in \mathbf{R}^3 : x^2 + y^2 + 2z = 2, z \geq 0\}$ , and let  $\hat{n}$  be the outward unit normal to  $S$ . If  $\vec{F} = y \hat{i} + xz \hat{j} + (x^2 + y^2) \hat{k}$ , then evaluate the integral  $\iint_S \vec{F} \cdot \hat{n} dS$ . (9)

(b) Let  $\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$  and  $r = |\vec{r}|$ . If a scalar field  $\varphi$  and a vector field  $\vec{u}$  satisfy  $\vec{\nabla} \varphi = \vec{\nabla} \times \vec{u} + f(r) \vec{r}$ , where  $f$  is an arbitrary differentiable function, then show that  $\nabla^2 \varphi = r f'(r) + 3f(r)$ . (6)



22. (a) Let  $D$  be the region bounded by the concentric spheres  $S_1: x^2 + y^2 + z^2 = a^2$  and  $S_2: x^2 + y^2 + z^2 = b^2$ , where  $a < b$ . Let  $\hat{n}$  be the unit normal to  $S_1$  directed away from the origin. If  $\nabla^2 \varphi = 0$  in  $D$  and  $\varphi = 0$  on  $S_2$ , then show that

$$\iiint_D |\vec{\nabla} \varphi|^2 dV + \iint_{S_1} \varphi (\vec{\nabla} \varphi) \cdot \hat{n} dS = 0. \quad (9)$$

- (b) Let  $C$  be the curve in  $\mathbf{R}^3$  given by  $x^2 + y^2 = a^2$ ,  $z = 0$  traced counter-clockwise, and let  $\vec{F} = x^2 y^3 \hat{i} + \hat{j} + z \hat{k}$ . Using Stokes' theorem, evaluate  $\oint_C \vec{F} \cdot d\vec{r}$ . (6)

23. Let  $V$  be the subspace of  $\mathbf{R}^4$  spanned by the vectors  $(1,0,1,2)$ ,  $(2,1,3,4)$  and  $(3,1,4,6)$ .  
Let  $T: V \rightarrow \mathbf{R}^2$  be a linear transformation given by  $T(x, y, z, t) = (x - y, z - t)$  for all  $(x, y, z, t) \in V$ . Find a basis for the null space of  $T$  and also a basis for the range space of  $T$ . (15)



24. (a) Compute the double integral  $\iint_D (x+2y) dx dy$ , where  $D$  is the region in the  $xy$ -plane bounded by the straight lines  $y=x+3$ ,  $y=x-3$ ,  $y=-2x+4$  and  $y=-2x-2$ . (9)

(b) Evaluate  $\int_0^{\pi/2} \left[ \int_{\pi/2}^{\pi} \frac{\sin x}{x} dx \right] dy + \int_{\pi/2}^{\pi} \left[ \int_y^{\pi} \frac{\sin x}{x} dx \right] dy$ . (6)

25. (a) Does the series  $\sum_{k=1}^{\infty} \frac{(-1)^k k + x^k}{k^2}$  converge uniformly for  $x \in [-1, 1]$ ? Justify. (9)

(b) Suppose  $(f_n)$  is a sequence of real-valued functions defined on  $\mathbf{R}$  and  $f$  is a real-valued function defined on  $\mathbf{R}$  such that  $|f_n(x) - f(x)| \leq |a_n|$  for all  $n \in \mathbf{N}$  and  $a_n \rightarrow 0$  as  $n \rightarrow \infty$ . Must the sequence  $(f_n)$  be uniformly convergent on  $\mathbf{R}$ ? Justify. (6)



26. (a) Suppose  $f$  is a real-valued thrice differentiable function defined on  $\mathbf{R}$  such that  $f'''(x) > 0$  for all  $x \in \mathbf{R}$ . Using Taylor's formula, show that

$$f(x_2) - f(x_1) > (x_2 - x_1) f' \left( \frac{x_1 + x_2}{2} \right) \text{ for all } x_1 \text{ and } x_2 \text{ in } \mathbf{R} \text{ with } x_2 > x_1. \quad (9)$$

(b) Let  $(a_n)$  and  $(b_n)$  be sequences of real numbers such that  $a_n \leq a_{n+1} \leq b_{n+1} \leq b_n$  for all  $n \in \mathbf{N}$ . Must there exist a real number  $x$  such that  $a_n \leq x \leq b_n$  for all  $n \in \mathbf{N}$ ? Justify your answer. (6)

27. Let  $G$  be the group of all  $2 \times 2$  matrices with real entries with respect to matrix multiplication. Let  $G_1$  be the smallest subgroup of  $G$  containing  $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  and  $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ , and  $G_2$  be the smallest subgroup of  $G$  containing  $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$  and  $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ . Determine all elements of  $G_1$  and find their orders. Determine all elements of  $G_2$  and find their orders. Does there exist a one-to-one homomorphism from  $G_1$  onto  $G_2$ ? Justify. (15)



28. (a) Let  $p$  be a prime number and let  $\mathbf{Z}$  be the ring of integers. If an ideal  $J$  of  $\mathbf{Z}$  contains the set  $p\mathbf{Z}$  properly, then show that  $J = \mathbf{Z}$ . (Here  $p\mathbf{Z} = \{px : x \in \mathbf{Z}\}$ .) (9)
- (b) Consider the ring  $R = \{a + ib : a, b \in \mathbf{Z}\}$  with usual addition and multiplication. Find all invertible elements of  $R$ . (6)

29. (a) Suppose  $E$  is a non-empty subset of  $\mathbf{R}$  which is bounded above, and let  $\alpha = \sup E$ . If  $E$  is closed, then show that  $\alpha \in E$ . If  $E$  is open, then show that  $\alpha \notin E$ . (9)

(b) Find all limit points of the set  $E = \left\{ n + \frac{1}{2m} : n, m \in \mathbf{N} \right\}$ . (6)



DO NOT WRITE ON THIS PAGE



|                                                                      |                  |
|----------------------------------------------------------------------|------------------|
| <b>2007 – MA</b><br><b>Objective Part</b><br><b>(Q. Nos. 1 – 15)</b> |                  |
| <b>Total Marks</b>                                                   | <b>Signature</b> |
|                                                                      |                  |

| <b>Subjective Part</b>                |              |                  |               |              |                  |
|---------------------------------------|--------------|------------------|---------------|--------------|------------------|
| <b>Q. No.</b>                         | <b>Marks</b> | <b>Signature</b> | <b>Q. No.</b> | <b>Marks</b> | <b>Signature</b> |
| 16                                    |              |                  | 23            |              |                  |
| 17                                    |              |                  | 24            |              |                  |
| 18                                    |              |                  | 25            |              |                  |
| 19                                    |              |                  | 26            |              |                  |
| 20                                    |              |                  | 27            |              |                  |
| 21                                    |              |                  | 28            |              |                  |
| 22                                    |              |                  | 29            |              |                  |
| <b>Total Marks in Subjective Part</b> |              |                  |               |              |                  |

|                                                |   |  |
|------------------------------------------------|---|--|
| <b>Total (Objective Part)</b>                  | : |  |
| <b>Total (Subjective Part)</b>                 | : |  |
| <b>Grand Total</b>                             | : |  |
| <b>Total Marks (in words)</b>                  | : |  |
| <b>Signature of Examiner(s)</b>                | : |  |
| <b>Signature of Head Examiner(s)</b>           | : |  |
| <b>Signature of Scrutinizer</b>                | : |  |
| <b>Signature of Chief Scrutinizer</b>          | : |  |
| <b>Signature of Coordinating Head Examiner</b> | : |  |



Examrace